# MANICHANDRA M R

**Software Engineer II | .NET & AI-Powered Claims Solutions**

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## PROFESSIONAL SUMMARY

Backend Software Engineer with 3+ years building and modernizing enterprise vehicle-claims platforms in C# / .NET at Solera. 4x Star of the Month awardee. Currently leading the cloud migration of a document-extraction service and building an Azure AI pipeline (Azure AI Document Intelligence + Azure OpenAI) that automates data extraction from estimate templates. Core strengths: REST API design, legacy modernization, high-volume file processing, production incident response, and AI-assisted developer tooling.

## TECHNICAL SKILLS

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| **Languages** | C#, SQL (T-SQL), JavaScript, HTML, CSS, XML, XSLT, PowerShell |
| **Backend (.NET)** | .NET, .NET Core, ASP.NET Core Web API, ASP.NET MVC, Entity Framework (Core), ADO.NET, LINQ, Dependency Injection, OOP, SOLID principles, design patterns, microservices, async/await, multi-threading & parallel processing, Hangfire (background jobs), Dapper, Windows Services, IIS |
| **APIs & Integration** | REST, JSON, SOAP/XML, Swagger/OpenAPI, Postman, OnBase (Hyland) |
| **AI / LLM** | Azure OpenAI Service (GPT models), Azure AI Document Intelligence (Form Recognizer), LLM-based data extraction, prompt engineering |
| **Database** | Microsoft SQL Server: stored procedures, query tuning, schema design |
| **Cloud & DevOps** | Microsoft Azure (AI services), Docker, Kubernetes (Rancher), GitHub Actions (CI/CD), Azure DevOps, HashiCorp Vault, S3 object storage, IIS deployment, PowerShell automation, Windows Task Scheduler, Git, GitHub |
| **Testing & Observability** | xUnit, NUnit, Moq, SonarQube, Playwright, Graylog, New Relic APM |
| **AI Developer Tooling** | GitHub Copilot, Claude Code, MCP servers (Graylog, S3, SQL Server, Jira, Confluence), custom Copilot skills & agents |
| **Practices** | Legacy modernization, production support (24-hr SLA), client onboarding, release management, Agile/Scrum, Jira, Confluence |

## PROFESSIONAL EXPERIENCE

### Solera Holdings — Bengaluru, India

Products: HyperQuest (vehicle claims management) and APD (auto physical damage assessment)

**Software Engineer II** — **Aug 2025 - Present**

- **Leading the cloud migration** of the document-extraction service. It is an **ASP.NET Core Web API** with **Hangfire** background jobs that pulls claim packages (BMS XML + PDF, TIFF, JPG, HTML, ZIP) from S3, converts pages to images, and uses **Azure OpenAI** to classify each page and extract header, totals, and line items into structured EMS data in SQL Server, reducing manual review.
- Improved extraction accuracy with **template detection** (Mitchell, CCC Pathways, Audatex) and layout hints, **consensus voting** for line-item extraction, VIN validation, and labour-hour and estimate/tow-bill **reconciliation**. Added GPT-5 reasoning-model support.
- Built the **Document Review STP evaluator**. It classifies estimates, tow bills, and ACVs, extracts their amounts, and either auto-completes the claim (**straight-through processing**) or routes it to human review with clear reasons. Also built claim-cancel evaluation and a **manual page-reprocess API**.
- Hardened reliability: **retries and a circuit breaker** for AI calls, CancellationToken propagation, **transactional** EMS saves, parallel AI calls (Task.WhenAll), reporting AI failures instead of a false "amount not found", and **liveness/readiness health checks** (SQL Server, S3, AI endpoint) for Kubernetes probes.
- Built **CI/CD pipelines in GitHub Actions** for QAE, CAE, and PRD with verified deployments and **one-click auto-rollback**. The service runs on Rancher Kubernetes with autoscaling, **Vault-managed secrets** (hardcoded credentials removed), and New Relic APM.
- Optimized prompts, OCR payloads, and **AI response caching** to reduce token usage per document. Built a reusable **S3 file-handling library**. The service runs as containerized (Docker) workloads on **Rancher-managed Kubernetes**, with unit tests in **xUnit/NUnit + Moq** and a passing SonarQube quality gate.
- Built **HQ Prompt Tester**, an internal application for testing AI extraction on page images and **Azure AI Document Intelligence** OCR output. It is used to optimize prompts by tracking **input, output, and total tokens**, and to compare **Azure OpenAI models**

side by side on extraction results, **accuracy, token usage, and speed**. Also built a failed-claims review feature to re-run failed extractions.

- Designed and built a **REST API integration** connecting OnBase (Hyland) with HyperQuest, **replacing 10 legacy VB scripts**, removing a third-party dependency from the document workflow, and **cutting cost by 30%**.
- Led the **migration of 72 Windows Servers** with a 4-engineer team, achieving **zero data loss** and near-zero downtime across the HyperQuest platform.
- Deliver APD features across the **full request flow** (client forms, .NET service endpoints, XSLT transforms, and SQL Server stored procedures) with carrier-specific configuration and isolation.
- Own **deployments**: deploy .NET applications and Windows Services to **IIS** in lower environments, and write **production deployment scripts and runbooks** for the SCM team, along with monthly **release branch** management.
- Resolved **critical (P1) production incidents** with a 24-hour SLA by live-debugging on NOC calls and proposing fixes during the call, minimizing user-facing impact for enterprise clients.
- Built a **Windows Service** that detects and quarantines corrupted PDF files before processing, eliminating a recurring class of downstream pipeline failures.
- Built **custom MCP servers** (Node.js) connecting GitHub Copilot to Graylog, S3 buckets, and multiple SQL Server environments, and integrated **Jira and Confluence** MCP servers, plus **custom Copilot skills and agents** for code review, documentation, and Jira analysis. This cut investigation time for the team.
- Drove technical **onboarding of 5 enterprise clients**, owning integration, configuration, and go-live support.
- Improved **logging and observability** (Graylog) across platform components, reducing troubleshooting time and mean time to resolution (MTTR).

**Associate Software Engineer** **May 2023 - Jul 2025**

- **Modernized a legacy Internet Explorer + ActiveX application** to run on all modern browsers and rebuilt its UI end to end, extending product life and improving performance and accessibility for every user.
- Re-engineered two Windows Services with **multi-threaded, parallel-session processing**, file-count tracking, and job-completion gates, cutting end-of-day batch processing time and increasing throughput.
- Built a fully **configurable data-retention engine** (company-level holds, claim-based filtering, claim-number extraction from filenames and XML) that safely automated cleanup of **9 TB of file-share data**, with CSV audit logs for every deletion.
- Automated Windows file-share storage maintenance with **PowerShell scripts** that purge old logs and work-area files, scheduled through **Windows Task Scheduler**.
- Migrated claim creation and outbound document delivery for **2 enterprise clients from legacy SOAP/XML to modern REST/JSON APIs**.
- Implemented **TXFB photo consolidation**, merging multiple claim images into a single outbound PDF to streamline client document delivery.

## ACHIEVEMENTS & RECOGNITION

- **4x Star of the Month** at Solera Holdings for delivery excellence.

## EDUCATION

**Bachelor of Engineering (B.E.), Electronics & Communication Engineering** **2018 - 2022**
GM Institute of Technology, Davangere, Karnataka (VTU)